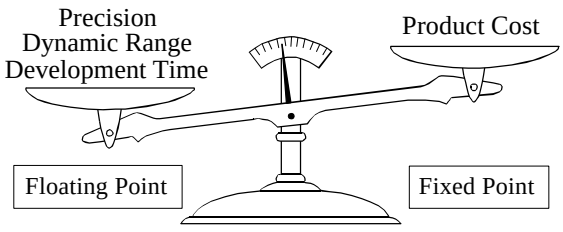

FIGURE 28-6
Fixed versus floating point. Fixed point DSPs are generally cheaper, while floating point devices have better precision, higher dynamic range, and a shorter development cycle.

faster than floating point in general purpose computers. However, with DSPs the speed is about the same, a result of the hardware being highly optimized for math operations. The internal architecture of a floating point DSP is more complicated than for a fixed point device. All the registers and data buses must be 32 bits wide instead of only 16; the multiplier and ALU must be able to quickly perform floating point arithmetic, the instruction set must be larger (so that they can handle both floating and fixed point numbers), and so on. Floating point (32 bit) has better precision and a higher dynamic range than fixed point (16 bit) . In addition, floating point programs often have a shorter development cycle, since the programmer doesn't generally need to worry about issues such as overflow, underflow, and round-off error.

On the other hand, fixed point DSPs have traditionally been cheaper than floating point devices. Nothing changes more rapidly than the price of electronics; anything you find in a book will be out-of-date before it is printed. Nevertheless, cost is a key factor in understanding how DSPs are evolving, and we need to give you a general idea. When this book was completed in 1999, fixed point DSPs sold for between $5 and $100, while floating point devices were in the range of $10 to $300. This difference in cost can be viewed as a measure of the relative complexity between the devices. If you want to find out what the prices are *today*, you need to look *today*.

Now let's turn our attention to *performance*; what can a 32-bit floating point system do that a 16-bit fixed point can't? The answer to this question is **signal-to-noise ratio**. Suppose we store a number in a 32 bit floating point format. As previously mentioned, the gap between this number and its adjacent neighbor is about one ten-millionth of the value of the number. To store the number, it must be round up or down by a maximum of one-half the gap size. In other words, each time we store a number in floating point notation, we add *noise* to the signal.

The same thing happens when a number is stored as a 16-bit fixed point value, except that the added noise is much worse. This is because the gaps between adjacent numbers are much larger. For instance, suppose we store the number 10,000 as a signed integer (running from -32,768 to 32,767). The gap between numbers is one ten-thousandth of the value of the number we are storing. If we